DRIVE

ON-ICE PERFORMANCE METRICS

The Nine Skating Metrics — And Why They Matter

Every metric below comes from wearable sensors worn during live play, so the numbers describe hockey as it actually happened — not a testing line. Read together, they answer three questions: **how fast is this player, how well do they change speed and direction,** and **how much can they physically produce and repeat.**

GROUP 01
SPEED & SKATING

Linear Speed

MEASURES
Top speed, average speed, and full speed distribution in a straight line.

The most visible separator on the ice — beating a defender wide, winning a race to a loose puck, running down a breakaway. Top speed alone is misleading: a player who touches 32 km/h once but sits at 14 km/h all shift plays slower than one who lives at 22. The **distribution** is what tells you which player you actually have.

Speed Bursts

MEASURES
Count and quality of short acceleration bursts, broken out by the speed they started from.

Shifts almost never last long enough to reach top speed. Hockey is decided in two- to four-second bursts. Sorting bursts by entry speed exposes whether a player can **re-accelerate while already moving** — the hidden skill behind puck retrievals, second-wave forechecks, and jumping into a rush late.

Agility

MEASURES
Top speed through turns, left-edge and right-edge top speeds, and agility acceleration force (a proxy for crossover power).

This is edge work. Comparing left to right exposes a weak side — almost every player has one, and opponents find it. Crossover force separates skaters who **generate speed inside a turn** from skaters who coast through it: escaping pressure on the wall, cutting through the corner, pivoting without giving up ground.

GROUP 02
TEMPO & TRANSITIONS

Pace

MEASURES
Percentage of ice time above 20 km/h, work-to-rest ratio, total skating distance, average distance per minute, and best distance per minute.

Pace is tempo, and it is the fairest way to compare unequal ice time: distance per minute puts a 40-second shift next to a 90-second one honestly. Work-to-rest shows whether a player is **driving play or recovering inside it** — two players with the same total distance can be having completely different games.

Acceleration

MEASURES
Bursts from under 5 km/h, distance gained during those bursts, sprints to full speed, top distance gained on a full sprint, and force distribution across accelerations.

The first three strides. In a fifteen-foot race — which is most races in hockey — getting off the mark beats top speed every time. **Distance gained** is the payoff: how much ice the player actually took. Force distribution says whether it came from a few powerful strides or many soft ones.

Deceleration

MEASURES
Peak deceleration, full stops from a range of entry speeds, delays (hard checks of speed that stop short of a standstill), and force distribution across decelerations.

Braking is what makes speed usable. A player who cannot stop hard has to turn wide, and arrives late and out of position. **Delays** are a tell for hockey sense — the puck carrier killing speed to buy a beat for support. These are also the highest-load actions a skater performs, which makes them worth watching.

GROUP 03
PHYSICAL OUTPUT

Strength

MEASURES
Single-leg and two-leg strength, right-turn and left-turn strength, and total g-force events through the legs.

Skating is force pushed into the ice, one leg at a time — which is why the single-leg number carries more weight than the two-leg number. A large **left-to-right gap** points to a compensation pattern that quietly caps top speed and raises injury risk. Total g-force events show how often the legs are truly being loaded.

Endurance

MEASURES
Total activity load (all accumulated g-force), load split across single leg, two legs, and left versus right, plus load broken out by force level.

Total load is the honest price of a session — more honest than ice time or shift count. **Where** the load sits explains soreness patterns and where breakdowns start. Splitting load by force level separates a lot of comfortable skating from genuinely hard work, and tells you what a player is prepared to absorb.

Conditioning

MEASURES
Top force against average force, all force events per minute, high-force events per minute, and total load per minute.

Conditioning here is not how long a player lasted — it is **how close to their own ceiling they stayed** from first shift to last. An average force sitting near peak means a repeatable player. High-force events per minute falling away by the third period means a fade. It is the difference between one good shift and twenty.